

- ① Find the distance between the line $\frac{x}{2} = \frac{2y-6}{4} = \frac{1-z}{-1}$ and another line parallel to it passing through the point $(4, 0, -5)$
- ② Using integration, find the area bounded by the ellipse $9x^2 + 25y^2 = 225$, the lines $x = -2$, $x = 2$ and the x -axis.
- ③ Given that $y = (\sin x)^x \cdot x^{\sin x} + a^x$, find $\frac{dy}{dx}$
- ④ Find the interval in which the function $f(x) = x^4 - 4x^3 + 10$ is strictly decreasing.
- ⑤ Express $\tan^{-1} \left(\frac{\cos x}{1 - \sin x} \right)$, where $-\frac{\pi}{2} < x < \frac{\pi}{2}$ in the simplest form.
- ⑥ Simplify: $\cos^{-1} x + \cos^{-1} \left(\frac{x}{2} + \sqrt{\frac{3-3x^2}{2}} \right)$; $\frac{1}{2} \leq x \leq 1$
- ⑦ If $y = (\tan^{-1} x)^2$, then show that $(x^2+1)^2 \frac{d^2y}{dx^2} + 2x(x^2+1) \frac{dy}{dx} = 2$
- ⑧ $\int \sin^{-1}(2x) dx$
- ⑨ $\int \frac{3x+5}{x^2+3x-18} dx$
- ⑩ $\int \frac{\sec^2 x dx}{\sqrt{\tan^2 x + 4}}$
- ⑪ $\int \frac{(x-3)e^x}{(x-1)^3} dx$